

LOGISTICS OPERATIONS OPTIMIZATION REPORT

A Data-Driven Analysis of Fleet Performance, Delivery Efficiency, and Maintenance Operations (2022–2024).

1. EXECUTIVE SUMMARY

This project presents a comprehensive analysis of logistics operations data from 2022 to 2024 with the objective of identifying operational inefficiencies, improving delivery performance, optimizing fleet utilization, and reducing maintenance-related costs.

The analysis was conducted primarily using SQL for data cleaning, transformation, feature engineering, and analytical querying, while Power BI was used exclusively for dashboard visualization and insight communication.

The company generated approximately **\$142.27 billion in total profit** across **82,061 trips** and covered over **117 million miles** during the study period. Despite this strong financial performance, several operational inefficiencies were identified, particularly in delivery reliability, maintenance management, route efficiency, and driver performance.

Key findings revealed that:

- Only **44.60% of deliveries** were completed on time
- More than **61% of shipments** experienced delays
- Certain routes generated high profit but also incurred excessive costs and delays
- Fleet utilization was strong overall but unevenly distributed
- Maintenance costs were significantly concentrated among heavily utilized trucks
- Driver performance varied substantially across operational metrics

The report concludes with strategic recommendations aimed at improving operational efficiency, reducing cost exposure, increasing customer satisfaction, and supporting long-term profitability.

2. INTRODUCTION

The logistics and transportation industry relies heavily on operational efficiency to maintain profitability and customer satisfaction. Companies must effectively manage transportation routes, fleet performance, delivery timelines, maintenance schedules, and driver productivity in order to remain competitive.

As logistics operations scale, operational complexity increases, making data-driven decision-making essential. This project leverages logistics operational data to evaluate business

performance and uncover actionable insights that can improve strategic and operational outcomes.

3. BUSINESS PROBLEM

The logistics company faces several operational challenges, including:

- Frequent delivery delays
- Rising maintenance costs
- Uneven fleet utilization
- Route inefficiencies
- Variations in driver performance

These challenges may negatively affect:

- customer retention,
- operational cost,
- asset lifespan,
- and overall profitability.

The purpose of this analysis is therefore to:

1. Evaluate operational performance
 2. Identify inefficiencies
 3. Discover profit drivers
 4. Recommend optimization strategies
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4. OBJECTIVES OF THE ANALYSIS

The primary objectives of the project are to:

- Analyze delivery performance across routes and drivers
- Evaluate profitability by truck, route, freight type, and customer type
- Assess fleet utilization and downtime patterns
- Investigate maintenance costs and operational impact
- Identify high-performing and underperforming drivers
- Provide data-driven operational recommendations

5. DATASET OVERVIEW

The dataset contains operational logistics records from 2022–2024 and includes the following tables:

- Trips
- Delivery events
- Drivers
- Driver monthly metrics
- Trucks
- Trailers
- Loads
- Safety incidents
- Fuel purchases
- Maintenance records
- Customers
- Facilities
- Routes
- Truck utilization metrics

The analysis covered:

- 82.06K trips
- 92 trucks
- 124 drivers
- 117 million miles travelled

6. METHODOLOGY

The analytical workflow followed the standard data analytics lifecycle:

6.1 Data Cleaning

SQL was used to:

- remove invalid records,

- handle missing values,
- validate joins,
- and standardize fields.

Examples included:

- filtering null distances,
 - removing negative costs,
 - validating timestamps.
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6.2 Data Transformation

Feature engineering was performed using SQL to derive:

- cost per mile,
 - revenue per mile,
 - average delay,
 - on-time delivery flags,
 - maintenance metrics,
 - and utilization measures.
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6.3 Exploratory Data Analysis

SQL queries were used to:

- aggregate route profitability,
 - analyze delivery delays,
 - assess fleet utilization,
 - and compare driver performance.
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6.4 Visualization

Power BI dashboards were developed for:

- Executive Overview
- Route Performance
- Fleet Utilization

- Driver Insights
 - Maintenance Analysis
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7. KEY PERFORMANCE INDICATORS (KPIs)

The analysis focused on the following KPIs:

KPI	Value
Total Profit	\$142.27bn
Total Cost	\$87.53bn
Total Trips	82.06K
Distance Covered	117M miles
On-Time Delivery Rate	44.60%
Average Delay Time	1.50 hours
Fleet Utilization Rate	83.13%
Total Maintenance Cost	\$3.87bn

8. ANALYSIS AND FINDINGS

8.1 DELIVERY PERFORMANCE ANALYSIS

Delivery performance emerged as the most significant operational concern.

The analysis showed:

- 61.04% of shipments were delivered late
- only 44.60% were delivered on time
- average delay time was 1.5 hours
- Deliveries to California, Washington, Texas, and Oregon accounted for over 30% of the late deliveries.

The most delay-prone routes included:

- Ohio → California
- Texas → Indiana
- Oregon → Washington

- Tennessee → Minnesota
- Nevada → New York

Ohio → California recorded:

- 946 late deliveries,
- with an average delay of 1.69 hours.

Interpretation

Persistent delays suggest:

- inefficient route scheduling,
- operational bottlenecks,
- fleet reliability issues,
- or poor resource allocation.

Business Impact

Poor delivery performance can result in:

- customer dissatisfaction,
- contract losses,
- and increased operational cost.

8.2 ROUTE PROFITABILITY ANALYSIS

Several routes generated exceptionally high profit levels.

Top-performing routes included:

- Ohio → Pennsylvania
- Pennsylvania → New York
- Ohio → California
- Nevada → California

However, some highly profitable routes also exhibited:

- extremely high cost per mile,
- and elevated delay frequency.

For example:

- Pennsylvania → New York recorded a cost per mile exceeding \$11,000 while remaining highly profitable.

Interpretation

This indicates that profitability may currently be sustained by high shipment volume rather than operational efficiency.

Business Risk

If operational costs continue increasing, these routes may become financially unstable.

8.3 FLEET UTILIZATION ANALYSIS

Fleet utilization was generally strong.

Key metrics included:

- Fleet utilization rate = 83.13%
- Average trips per truck = 892

Volvo trucks contributed the highest operational output:

- Highest trips completed
- Highest profit generated

However, utilization appeared uneven across truck categories.

Interpretation

Overreliance on certain trucks increases:

- maintenance exposure,
 - wear and tear,
 - and operational dependency.
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8.4 DRIVER PERFORMANCE ANALYSIS

Driver performance varied significantly.

Top profit-generating drivers included:

- William Wilson
- Thomas Gonzalez
- John Thomas

Top on-time drivers included:

- Joseph Martinez
- Robert Brown
- Thomas Wilson

The analysis also showed that:

- driver experience positively correlated with on-time delivery performance.

Interpretation

Operational productivity alone is not sufficient for evaluating drivers.

Some drivers generate high revenue while still contributing to delays.

Recommendation

A composite driver performance score should combine:

- punctuality,
 - profitability,
 - safety,
 - and fuel efficiency.
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8.5 MAINTENANCE ANALYSIS

Maintenance represented a major operational cost area.

Key findings included:

- Total maintenance cost = \$3.87bn
- Total maintenance events = 2,920
- Average downtime = 13.62 hours

Volvo trucks recorded the highest maintenance cost at approximately \$97M.

Maintenance categories were relatively evenly distributed across:

- inspection,
- preventive maintenance,
- engine,
- repair,
- brake,
- tire,

- and transmission services.

Interpretation

The issue appears to be broad fleet wear rather than isolated mechanical failures.

Business Impact

High downtime negatively affects:

- delivery performance,
 - fleet availability,
 - and operational efficiency.
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9. STRATEGIC RECOMMENDATIONS

Based on the analysis, the following recommendations are proposed:

9.1 Improve Delivery Scheduling

The company should:

- optimize scheduling for high-delay routes,
- revise dispatch timing,
- and implement route optimization models.

9.2 Implement Predictive Maintenance

Predictive maintenance should be introduced using:

- mileage,
- maintenance history,
- and downtime patterns.

This would reduce:

- unexpected failures,
- downtime,
- and maintenance cost.

9.3 Develop a Driver Performance Scoring System

A weighted driver score should incorporate:

- on-time delivery rate,
- profitability,
- safety performance,
- fuel efficiency,
- and delay frequency.

9.4 Balance Fleet Utilization

Truck assignments should be distributed more evenly to:

- reduce overuse,
- minimize maintenance pressure,
- and extend asset lifespan.

9.5 Monitor Route-Level Profitability

The company should continuously track:

- route profit,
- cost per mile,
- delay frequency,
- and maintenance exposure.

9.6 Protect High-Value Customers

High-revenue customers should receive:

- priority operational attention,
- improved service reliability,
- and proactive communication.

10. CONCLUSION

This project demonstrated how logistics operational data can be transformed into actionable business intelligence using SQL-driven analytics and Power BI visualization.

Although the company is highly profitable, the analysis revealed critical operational inefficiencies in:

- delivery reliability,
- maintenance management,
- route optimization,
- and fleet balancing.

Addressing these issues can significantly improve:

- customer satisfaction,
- operational efficiency,
- asset longevity,
- and long-term profitability.

The project also demonstrates the value of combining:

- SQL for analytical processing,
- and Power BI for visual storytelling,

in solving real-world logistics problems.